A Lightweight Noise-Robust Post-Trigger Classifier for Embedded Acoustic Artillery Launch Detection

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***Abstract*—Acoustic artillery sensing in the field must tolerate background noise while running on low-power embedded hardware. We study the post-trigger classification stage of a lightweight two-stage pipeline for artillery launch detection. The corpus contains 841 labeled events: 50 artillery launch events, each recorded by four synchronized microphones, and 791 non-launch events. We train the classifier with an expanded non-launch corpus, class-imbalance mitigation, and train-time augmentation based on additive waveform noise; a controlled comparison shows that mel-frequency cepstral coefficient (MFCC)-domain perturbation is not a substitute for it. We evaluate the classifier on a held-out test partition under waveform-domain additive noise applied before MFCC extraction at acoustic signal-to-noise ratio (SNR) levels of 30, 20, 10, and 5 dB, with stress levels of 0 and -5 dB. The embedded experiment validates only the post-trigger classifier: deployed on ESP32-S3 with TensorFlow Lite Micro, it preserves high performance under the tested conditions, with Matthews correlation coefficient (MCC) values of 0.99 at 30 dB and 0.98 at 5 dB. The results support the use of a lightweight embedded post-trigger classifier as part of a two-stage detection pipeline.**

***Index Terms*—acoustic detection, artillery launch, MFCC, CNN, data augmentation, TensorFlow Lite, embedded AI, robustness, waveform-domain noise**

I. Introduction

Passive acoustic sensing can support early warning in contested environments where active systems are expensive, power-demanding, or easier to locate. Recent high-intensity conflicts have increased the need for low-cost passive systems that can detect artillery launches without emitting signals of their own. A practical acoustic system for this task must satisfy two constraints: (i) robustness to background noise and event variability, and (ii) feasible execution on low-power embedded hardware.

Existing acoustic weapon-detection pipelines usually separate continuous-event triggering from the interpretation of short post-trigger signal windows. Triggering often relies on thresholding, energy-envelope analysis, time-of-arrival logic, or multi-sensor consistency checks. Post-trigger classification then uses engineered time-frequency or cepstral descriptors [1]–[3] and, more recently, convolutional neural network (CNN)-style classifiers [4], [5]. For embedded artillery-launch sensing, the key question is whether efficient mel-frequency cepstral coefficient (MFCC)-domain training augmentation can substitute for waveform-domain noise augmentation when a compact classifier faces waveform-domain additive noise and quantized deployment on low-power hardware.

We adopt a two-stage processing architecture. A streaming trigger first identifies candidate impulsive events, and a post-trigger classifier then processes short acoustic segments. We treat the first stage as previously validated prior work and focus on the artillery launch classifier. Within this scope, we test whether a lightweight classifier preserves useful discrimination under controlled waveform-domain additive noise reported as acoustic signal-to-noise ratio (SNR) levels, while remaining deployable on ESP32-S3.

Our training strategy combines an expanded non-launch corpus, class-imbalance mitigation, and train-time waveform-domain noise augmentation, selected by a controlled comparison against MFCC-domain augmentation. We also assess whether the same classifier remains useful after quantization and deployment through TensorFlow Lite Micro on ESP32-S3, using PC-side 8-bit integer (int8) inference as the immediate reference point.

II. State of the Art

A. Classical Acoustic Detection of Impulsive Events

Weapon-related acoustic surveillance has traditionally used thresholding, energy-envelope analysis, time-of-arrival logic, and multi-sensor consistency checks to detect impulsive transients before semantic classification. Operational pipelines usually split artillery and gunshot processing into two tasks: (i) event triggering in continuous audio streams and (ii) post-trigger interpretation of short signal segments. Battlefield-oriented studies show that reliable operation depends on tolerating propagation distortion, changing stand-off distances, and heterogeneous sensor layouts [1], [2], [6], [7]. Trigger quality therefore limits downstream classifier performance.

B. Acoustic Signature of Artillery and Gunfire

Artillery and gunfire recordings do not follow a single invariant waveform. Source type, firing geometry, propagation, and local reflections shape each event. Depending on scenario and range, a recording may include different mixtures of muzzle blast, shock waves from supersonic projectiles, and delayed arrivals. Fig. 1(a) shows an example artillery-launch recording.

Prior studies captured these effects indirectly through time-frequency or cepstral descriptors. Long-range studies modeled wave arrivals explicitly and documented sensitivity to atmosphere, terrain, and sensor geometry [7].

C. From Detection to Post-Detection Classification

After trigger detection, systems usually classify short post-trigger windows. Classical approaches combine engineered representations, such as wavelet, cepstral, or temporal-envelope features, with shallow classifiers, reaching, for example, 97.5 % launch/impact discrimination and 93.6 % variant classification of mortar and artillery rounds in single-condition evaluations [2], [6]. Later methods use CNN models over spectro-temporal inputs [4], [8], with reported firearm-identification accuracies above 90 % [5]. Cepstral representations, introduced for speech recognition [9], remain attractive because they compactly encode the spectral envelope of short signal segments. Comparative evidence from related gunshot-classification experiments suggests that performance depends on the interaction between the feature family and the classifier, not on model choice alone. In those comparisons, gammatone-based features outperformed MFCC variants [3]; we nevertheless adopt MFCCs as a compact, widely standardized front end with mature embedded implementations. These findings favor a two-stage design with a robust trigger and a compact post-trigger classifier. Lightweight anomaly detectors (one-class support vector machines, isolation forests, autoencoders) flag deviations from a known acoustic background. After the trigger, however, every candidate is already anomalous, including non-artillery gunshots, so we train a supervised binary classifier on labeled events instead.

These considerations motivate compact CNN topologies that preserve local time-frequency structure while remaining deployable on constrained hardware. A small stack of local convolutions, pooling, and a dense classification head follows the established LeNet-like pattern for compact two-dimensional inputs [10]. In audio classification, analogous CNNs operate on time-frequency representations and learn local spectro-temporal patterns rather than hand-coded feature interactions [4], [11]. For microcontroller-oriented deployments, this small-footprint design logic is often preferable to deeper architectures when memory, latency, and operator support are limiting factors [12]. In this work, we use a task-specific compact MFCC-CNN adapted to the $(58, 12)$ input geometry and validate it as the second stage of the embedded artillery-launch pipeline.

D. Robustness and Domain-Shift Handling

Recent studies report strong performance under uncontrolled recording conditions, but many results remain tied to dataset-specific priors, acquisition pipelines, and label granularity [3], [5], [13]. In practice, model performance can degrade under perturbation, recording-device mismatch, or changes in background scenes, and overlapping sound sources further disrupt classification [14]. Clean-set accuracy alone is therefore insufficient. Recent embedded-oriented firearm classification fuses mel-spectrogram, MFCC, and linear frequency cepstral coefficient (LFCC) inputs in a convolutional recurrent network, reporting 96.3 % accuracy with 187 ms inference per 2 s segment on application-class hardware (Raspberry Pi 4) [15]; a systematic robustness characterization across controlled SNR levels and a microcontroller unit (MCU)-class deployment remain open. Augmentation strategy, perturbation tolerance, and operating-point calibration become central requirements [8]. For embedded artillery-launch sensing, the relevant test is whether class separability survives waveform-domain additive noise before feature extraction, because this more closely follows the acoustic front end encountered by a deployed system.

Feature-domain augmentation has methodological precedent: SpecAugment applies warping and masking directly to log-mel feature inputs rather than to raw audio [16]. In noise-robust automatic speech recognition (ASR), feature-level Gaussian injection has been compared directly with waveform-level noise mixing, and the waveform-level variant transferred robustness better [17]. Moreover, waveform-level additive noise and cepstral-domain perturbations are not equivalent: additive noise before feature extraction produces nonlinear, signal-dependent distortions in cepstral space [18]. Whether cepstral-domain perturbations can substitute for waveform-domain corruption in short impulsive-event classification is therefore an open question.

E. Embedded Deployment and Runtime Constraints

Edge deployment requires detection quality to be evaluated together with memory footprint, latency, and energy use. Tiny machine learning (TinyML) comparisons show that front-end feature extraction can dominate runtime, so the highest-accuracy model is not always the most deployable one [19]. MCU-focused TinyML systems also show that memory, latency, energy, and runtime overhead can dominate architecture and deployment choices on commodity microcontrollers [12], [20]. Together with TensorFlow Lite deployment practice [21], [22] and integer or fixed-point quantization methods for efficient on-device inference [23], [24], these findings motivate validation that separates desktop float32 behavior, quantized desktop verification, and embedded execution.

This work addresses two specific questions: (i) whether robustness of a compact post-trigger artillery classifier to uncorrelated additive acoustic noise requires training-time perturbation of the waveform, or whether perturbing the cepstral features is sufficient, and (ii) whether the resulting robustness survives int8 quantization and MCU-class deployment. Both questions are answered experimentally.

III. Method

A. Processing Pipeline

In the full architecture, the first stage follows a median-filter impulse-detection approach: a streaming trigger estimates the local acoustic background, identifies candidate impulsive events, and forwards short buffered segments to a classifier.

The trigger, acquisition chain, and online buffering logic are outside the scope of this study; their design and validation are reported in prior work, where the median-filter trigger detected all tested small-arms gunshots in shooting-range measurements and operated reliably down to an acoustic SNR of 5 dB [25]. We evaluate only the second stage, the post-trigger CNN artillery launch classifier.

Each candidate event is represented by a 60 ms waveform segment centered on the detected impulse peak. The segment is asymmetric: 30 % precedes the peak to capture onset structure, and 70 % follows it to retain decay and muzzle-blast content. Fig. 1 shows an example launch recording (a) and the resulting analysis window (b). We then compute MFCC features from these segments for supervised CNN training and inference.

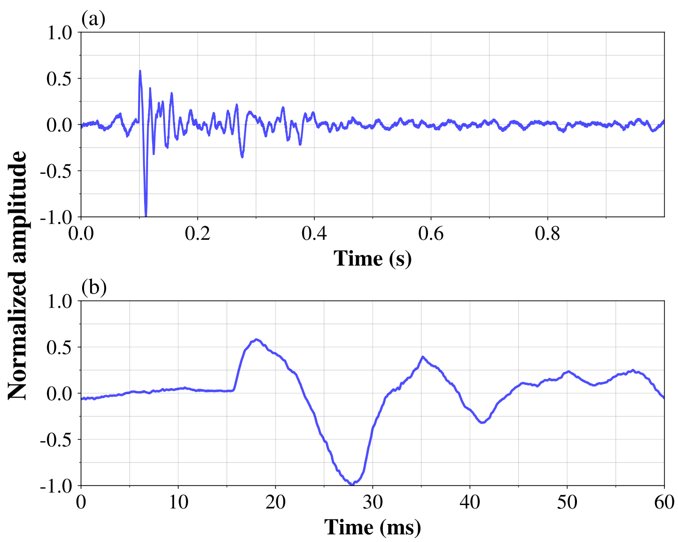

Fig. 1. (a) Example artillery-launch recording; (b) the 60 ms peak-centered analysis window extracted by the processing pipeline.

B. Field Data Acquisition

The source corpus comes from real artillery measurements and includes both launch and impact acoustic events. We collected the recordings with the field station shown in Fig. 2. The acquisition chain used PreSonus PRM1 microphones connected through a Roland Rubix 44 audio interface, and the signal processing uses 16-bit audio at 22.05 kHz. We model only binary artillery launch classification (artillery launch vs. non-launch) and leave impact-event modeling for future work. The positive class contains only artillery launch events, whereas the non-launch class contains impulsive false-trigger candidates, including door impacts, human shouts, bell sounds, small-arms gunshots, and other non-artillery recordings.

The clean corpus (Table I) combines 50 physical launch events, recorded on four synchronized microphone channels over two measurement days with a single 152 mm self-propelled howitzer, with 706 impulsive-noise events and 85 small-arms gunshots as hard negatives. The recording station stood 1–2 km from the impact area; the guns fired from an estimated 5–10 km from the station. The split is performed at the event level, so channels of one shot never cross partitions: training uses all four channels per launch event, raising the launch share of the training samples to 20 % (an effective imbalance of 1:4, matched by the class-weighted loss [26]), while validation and test use one channel per event. The small number of launch events remains the dominant source of statistical uncertainty (Section VI).

TABLE I
LABELED CORPUS AND EVENT-LEVEL SPLIT. TRAINING USES ALL FOUR MICROPHONE CHANNELS PER LAUNCH EVENT; VALIDATION AND TEST USE ONE.

Subset	Events	Samples Train	Samples Val.	Samples Test
Artillery launch (152 mm howitzer)	50	128	8	10
Non-launch: impulsive noise	706	457	109	140
Non-launch: small-arms gunshots	85	49	18	18
Total corpus	841	634	135	168

Fig. 2. Field recording station used for artillery dataset acquisition.

C. CNN Topology

The network input is an MFCC tensor of size $(58, 12, 1)$ in `channels_last` format. The classifier first applies a valid Conv2D layer with 32 filters, a 3×3 kernel, stride 1, and rectified linear unit (ReLU) activation, producing a $(56, 10, 32)$ tensor. A 2×2 MaxPooling2D layer with stride 2 reduces this representation to $(28, 5, 32)$. The second valid Conv2D layer uses 64 filters with the same 3×3 kernel, stride 1, and ReLU activation, producing $(26, 3, 64)$. A second 2×2 MaxPooling2D layer with stride 2 then yields $(13, 1, 64)$. The result is flattened to 832 features, passed through Dense(64) with ReLU, regularized by Dropout(0.5), and classified by a final Dense(1) sigmoid output. The trainable parameter counts are 320 for the first convolution, 18,496 for the second convolution, 53,312 for Dense(64), and 65 for the output layer, for a total of 72,193 trainable parameters; the pooling, flattening, and dropout layers add no trainable parameters. Fig. 3 shows the full architecture. We train the model for up to 50 epochs with the Adam optimizer and binary cross-entropy loss, with early stopping on the validation loss restoring the best weights. This layout balances representation capacity and deployment cost, following standard convolutional design practice [10].

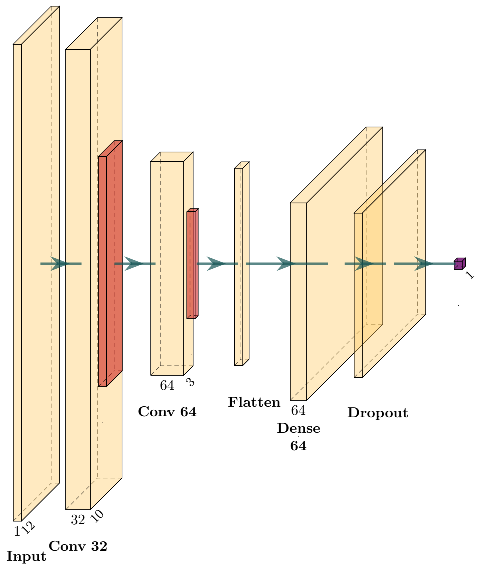

Fig. 3. CNN topology used for post-trigger artillery launch classification of MFCC segments.

D. Robustness Training

We extract MFCCs from 22.05 kHz signals using short-time framing with high temporal resolution. The rate is a deliberate choice for the embedded target: audio buffers and MFCC compute scale with the sampling rate, while the dominant energy of the launch signature lies well below the resulting 11 kHz Nyquist limit, so higher rates would spend scarce on-chip RAM without adding usable signal content. Recordings from heterogeneous sources (native rates up to 48 kHz) are resampled to this common rate. Each 60 ms segment is divided into 2.5 ms frames (approximately 55 samples) with a 1.0 ms hop (approximately 22 samples), corresponding to 1.5 ms overlap (60 %). A Hamming window is applied to each frame to reduce spectral leakage, and the spectrum is evaluated with a 512-point fast Fourier transform ($NFFT = 512$). The mel front end uses 40 mel filters, from which the first 12 MFCC coefficients are retained. This yields the $(58, 12, 1)$ input tensor used by the classifier.

The training strategy combines three elements: (i) an expanded non-launch corpus, (ii) class-imbalance mitigation, and (iii) train-time augmentation of the training split only. This follows the broader use of augmentation for audio classification under limited-data conditions [8]. We compare two augmentation domains: Gaussian perturbations applied directly to the precomputed MFCC matrices as a feature-space input-noise regularization strategy, and additive Gaussian noise applied to the training waveforms before MFCC extraction at SNR levels of 30, 20, 10, and 5 dB. Since the CNN receives MFCC tensors as its input, the MFCC-domain variant acts on the representation seen by the network and encourages stable predictions for locally perturbed cepstral representations. Training with input noise is theoretically related to Tikhonov regularization [27]. The deployed classifier is trained with the waveform-domain augmentation, selected by the comparison in Section V. The robustness results below use a separate waveform-domain evaluation protocol in which additive Gaussian noise, with noise realizations disjoint from the training augmentation, is applied to the audio signal before MFCC extraction.

E. Embedded Inference Protocol

After PC training, we export the selected model to TensorFlow Lite Micro [21], [22]. To separate hardware effects from quantization effects, we evaluate three inference regimes: desktop float32 inference, desktop int8 inference with the quantized model, and ESP32-S3 deployment of the same quantized classifier. The embedded implementation uses integer-only int8 inference [23] on ESP32-S3 with a statically allocated 80 KiB `tensor_arena` and a selective `MicroMutableOpResolver` that registers ten operators covering the eight used by the network, all supported natively by TensorFlow Lite Micro: the compute-intensive ones run through ESP-NN accelerated kernels, and the rest are lightweight data-movement operations. The final int8 model occupies 81,400 bytes (72,193 parameters); the firmware uses 201.5 kB of flash code, 144.0 kB of flash data, and 143.8 kB of RAM, dominated by the `tensor_arena`. Embedded validation uses a host-device protocol: the host generates waveform-domain noisy test variants, computes MFCC tensors from those modified waveforms, transmits the tensors to the embedded target, and collects predicted labels and scores. The ESP32-S3 experiment therefore validates the deployed TFLite Micro classifier only; the reported latency, measured on-device by the microsecond system timer immediately around the interpreter invocation, corresponds to classifier inference on one MFCC segment, not to the complete audio acquisition, triggering, feature extraction, and classification pipeline. We use ESP32-S3 as a representative low-power Edge AI platform for classifier deployment, not as a proof of universal generalization across embedded hardware.

IV. Experimental Setup

The corpus is split at the event level into training, validation, and test partitions (Table I); the validation partition drives early stopping and model selection, and the held-out test partition is used only for the final evaluation. Training is fully seeded, and every reported value is the mean ± std over five training seeds and five noise-realization seeds (25 runs per table cell). The acoustic SNR is defined per event: zero-mean Gaussian noise with variance $\sigma_n^2 = P_x \cdot 10^{-\mathrm{SNR}/10}$, where P_x is the mean power of the clean 1 s event clip, is added to the entire waveform before MFCC extraction at SNR levels of 30, 20, 10, and 5 dB, with additional stress levels of 0 and -5 dB beyond the validated trigger envelope; peak detection and window extraction are re-run on the noisy signal, so noise-induced window-placement errors are included. We report accuracy, precision, recall, F1-score, and Matthews correlation coefficient (MCC) using a fixed sigmoid decision threshold of 0.5, and evaluate each model as float32 inference and as int8 inference after full-integer quantization; the selected deployed model is additionally validated as int8 inference on ESP32-S3.

V. Results

A. Comparison of Training Augmentation Strategies

To answer the first question from Section II, we trained the same network three times and changed only how the training data were augmented: (i) Gaussian jitter added directly to the MFCC features, (ii) the same jitter combined with additive waveform noise, and (iii) additive waveform noise alone. Table II compares the three variants on the held-out test set. Jitter in the feature space fails at low SNR (int8 MCC 0.30 ± 0.21 at 5 dB, individual training seeds between 0.05 and 0.48), whereas waveform noise alone maintains performance across the entire tested range (0.98 ± 0.04 at 5 dB). Adding jitter on top of waveform noise makes the results slightly worse at every SNR level, so perturbing the cepstral features is not only insufficient but counterproductive; this contrasts with noise-robust ASR, where feature-level injection complemented waveform-level mixing [17]. The jitter-trained models also lose more when quantized (float32 MCC 0.37 versus int8 0.30 at 5 dB), whereas the waveform-trained models quantize without loss.

TABLE II
HELD-OUT INT8 MCC (MEAN ± STD) FOR THE SAME ARCHITECTURE TRAINED WITH THREE AUGMENTATION STRATEGIES.

Training augmentation	Clean	30 dB	20 dB	10 dB	5 dB
MFCC-domain jitter	0.82 ± 0.09	0.77 ± 0.11	0.56 ± 0.05	0.37 ± 0.18	0.30 ± 0.21
Jitter + waveform noise	0.97 ± 0.03	0.95 ± 0.04	0.92 ± 0.07	0.85 ± 0.08	0.86 ± 0.07
Waveform noise only	0.99 ± 0.02	0.99 ± 0.02	0.99 ± 0.02	0.98 ± 0.04	0.98 ± 0.04

B. Held-Out Robustness of the Deployed Classifier

Table III reports the held-out performance of the deployed waveform-augmented classifier. Performance is nearly flat across the trained range, with int8 MCC between 0.976 and 0.992 from clean conditions down to 5 dB. Beyond the validated trigger envelope, the classifier degrades gracefully (0.947 at 0 dB and 0.833 at -5 dB). The failure mode is operationally benign: precision stays at 1.000 from 5 dB downward while recall decreases, so under extreme noise the classifier misses launches rather than raising false alarms; across all noise levels, no small-arms gunshot in the test set was ever classified as an artillery launch. The float32 column shows the same models before quantization: the differences stay within ± 0.02 MCC without a consistent direction, so int8 quantization is performance-neutral for this classifier.

C. ESP32-S3 Deployment Results

The deployed model was verified on the target hardware by transmitting all held-out MFCC tensors to the ESP32-S3 and comparing its predictions with PC-side int8 inference. Of 1,176 on-device inferences across all seven noise levels,

TABLE III
HELD-OUT ROBUSTNESS OF THE DEPLOYED INT8 CLASSIFIER (MEAN ± STD OVER FIVE TRAINING AND FIVE NOISE SEEDS).

SNR (dB)	Accuracy (%)	Precision	Recall	F1-score	MCC	MCC (float32)
Clean	99.88 ± 0.24	0.982 ± 0.037	1.000 ± 0.000	0.990 ± 0.019	0.990 ± 0.020	1.000 ± 0.000
30	99.86 ± 0.26	0.982 ± 0.037	0.996 ± 0.020	0.988 ± 0.021	0.988 ± 0.022	0.998 ± 0.011
20	99.90 ± 0.22	0.996 ± 0.018	0.988 ± 0.033	0.992 ± 0.019	0.992 ± 0.020	0.993 ± 0.018
10	99.74 ± 0.39	0.985 ± 0.034	0.972 ± 0.061	0.977 ± 0.035	0.977 ± 0.035	0.974 ± 0.039
5	99.74 ± 0.42	1.000 ± 0.000	0.956 ± 0.071	0.976 ± 0.039	0.976 ± 0.039	0.978 ± 0.036
0[a]	99.43 ± 0.53	1.000 ± 0.000	0.904 ± 0.089	0.947 ± 0.049	0.947 ± 0.049	0.934 ± 0.045
-5[a]	98.31 ± 1.01	1.000 ± 0.000	0.716 ± 0.170	0.823 ± 0.122	0.833 ± 0.107	0.820 ± 0.088

[a]Beyond the trigger stage's validated 5 dB envelope (Section III-A).

1,171 were bit-exact; the five disagreements are decision-boundary samples whose PC-side score equals exactly 0.5, where the coarse logit quantization of this confident model (one least-significant bit spans 10.8 logit units) amplifies a legal one-bit rounding difference between the TFLite and TFLite Micro kernels. Platform behavior is therefore equivalent up to rounding at the decision boundary. The classifier inference latency per MFCC segment is 32.0 ms with sub-0.1 ms variation across the 1,176 inferences (range 31.99–32.09 ms), independent of the noise level. The quantized post-trigger classifier remains deployable on the target platform under the host-device validation protocol.

VI. Discussion and Limitations

Because each classified segment is centered on a trigger-detected peak, the reported figures characterize classifier performance under the assumption of successful impulse localization by the previously validated first stage [25]. The evaluated SNR range of the classifier deliberately matches the trigger stage's validated 5 dB envelope; the 0 and -5 dB stress levels probe classifier margin beyond the range in which the full two-stage system is expected to operate.

The study has several limitations. We do not evaluate continuous-stream trigger performance or false-trigger rates, so the system-level claim is limited to post-trigger classification. Model development used a single stratified event-level split, so the results quantify perturbation stability under that split rather than repeated-split or session-independent generalization. The noise study uses synthetic additive Gaussian noise in the waveform domain; this is closer to the deployed acoustic front end than MFCC-domain perturbation, but it still omits many field-noise, propagation, and recording-device shifts. The corpus also remains small in positive artillery-launch events and covers a single artillery type, so generalization across calibers is untested.

VII. Conclusion

We presented a lightweight two-stage acoustic pipeline for embedded artillery launch detection and evaluated its post-trigger classification stage under waveform-domain additive noise applied before MFCC extraction.

On the expanded 841-event corpus, MFCC-domain training augmentation leaves the classifier vulnerable to waveform-domain noise, whereas waveform-domain noise augmentation keeps it robust under the tested SNR levels, with quantization remaining performance-neutral. PC-side int8 verification and ESP32-S3 deployment show that the post-trigger classifier can run on the target platform under the host-device validation protocol, reaching MCC values of 0.99 at 30 dB and 0.98 at 5 dB with approximately 32 ms average classifier inference latency per MFCC segment and an 80 KiB `tensor_arena`.

The main contributions are the controlled comparison of training augmentation domains and the embedded validation of a practical second-stage classifier built around a task-specific lightweight MFCC-CNN under waveform-domain perturbations. Future work should extend the validation to continuous-stream false-trigger rates, on-device MFCC extraction, end-to-end latency, power and energy per classified decision, more diverse field-noise conditions, repeated-split or session-independent uncertainty estimation, and broader deployment targets such as STM32-class MCUs.

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